University course pack

# Teaching optimisation at university

A 14-week, 3-credit module in optimisation and operations research for third-year engineering, computer science or mathematics students. Prerequisites: linear algebra, calculus and programming. The four Darsi lectures anchor weeks 2, 6, 9 and 11.

## Learning outcomes

1. Formulate linear and integer programs from realistic briefs.
2. Solve small LPs geometrically and interpret duals and sensitivity.
3. Explain and apply branch and bound and formulation strength.
4. Implement and fairly evaluate local search and simulated annealing.
5. Validate models with stakeholders and communicate results.

## 14-week syllabus

| Week | Topic | In class and out |
|---|---|---|
| 1 | What is optimisation? Models and decisions | Open brief 1 |
| 2 | Linear programming and geometry | Lab 1: draw it, then price it |
| 3 | The simplex method | Problem set 1 |
| 4 | Duality and sensitivity | Solver memo assignment |
| 5 | Network flows and transport | Problem set 2 |
| 6 | Integer programming | Lab 2: branch and bound by hand |
| 7 | Formulation strength and cuts | Midterm exam |
| 8 | Dynamic programming | Problem set 3 |
| 9 | Local search and annealing | Lab 3: every row a seed |
| 10 | Other metaheuristics | Heuristics benchmark |
| 11 | The modelling cycle | Project start: open brief |
| 12 | Nonlinear and convex optimisation | Problem set 4 |
| 13 | Optimisation under uncertainty | Project checkpoint |
| 14 | Project defences and review | Final exam |

University course pack

# Lecture slide outline

**Lecture 1: Linear programming**

- The canteen menu
- Feasible regions and vertices
- The simplex idea
- Duality and shadow prices
- Activity: draw it, then price it

**Lecture 2: Integer programming**

- Two and a half buses
- Binary decisions
- Relaxations as bounds
- Branch, bound, prune
- Activity: grow the tree

**Lecture 3: Heuristics**

- Six hundred exams
- Local search and local optima
- Simulated annealing
- Fair evaluation
- Activity: every row a seed

**Lecture 4: Modelling**

- The right answer to the wrong problem
- The modelling cycle
- Open briefs and stakeholders
- Assessing models
- Activity: interview, formulate, defend

University course pack

# Problem set

**1.** Solve max $5x + 4y$ s.t. $6x + 4y \le 24$, $x + 2y \le 6$, $x, y \ge 0$ graphically.

**2.** Write the dual of the problem above and verify strong duality.

**3.** Knapsack: items (weight, value) (3, 8), (4, 9), (5, 11), (2, 4); capacity 9. Solve by branch and bound.

**4.** Model a fixed cost: opening facility i costs $f_i$ and allows shipping up to $U_i$. Write the constraints.

**5.** Annealing starts at $T = 10$ with geometric cooling $\alpha = 0.9$. After how many steps is T below 1, and what is then the acceptance probability for $\Delta = 1$?

**6.** List four questions you would ask a stakeholder before modelling a campus parking allocation.

## Solutions

**1.** Vertices (0,0), (4,0), (3,1.5), (0,3) give 0, 20, 21, 12. The optimum is 21 at (3, 1.5).

**2.** min $24u + 6v$ s.t. $6u + v \ge 5$, $4u + 2v \ge 4$, $u, v \ge 0$. Solution $u = 0.75$, $v = 0.5$ gives $18 + 3 = 21$, equal to the primal.

**3.** The relaxation gives $8 + 9 + (2/5) \times 11 = 21.4$. Branching on item 3 finds items 1, 2 and 4 (weight 9, value 21); since every integer value is whole and the bound is 21.4, no solution can exceed 21, so 21 is optimal.

**4.** Binary $y_i$; objective includes $\Sigma\, f_i\, y_i$; shipping $\Sigma_j\, x_{ij} \le U_i\, y_i$, so nothing ships from a closed facility.

**5.** $10 \times 0.9^k < 1$ needs $k > \ln 10 / \ln(1/0.9) \approx 21.9$, so $k = 22$, $T \approx 0.98$, and $p = e^{(-1/0.98)} \approx 0.36$.

**6.** Who must be guaranteed a space? What does fairness mean here? Which times and days matter? What may not change, such as accessible spaces?

# Lab handout: Lab 3: Every row a seed

## Goal

Compare local search and simulated annealing on exam timetabling and evaluate them fairly.

## Steps

1. Load the training and test timetabling instances.
2. Implement the move: reassign one exam.
3. Tune the cooling rate on training instances only.
4. Run both methods with ten seeds and equal time budgets on test instances.
5. Plot distributions and cost against time; report gaps to known optima.

## Hand in

Code, plots and a two-page evaluation report.

University course pack

# Assessment rubric

| Criterion | Excellent | Good | Developing | Not yet |
|---|---|---|---|---|
| **Formulation** | Correct, strong and justified | Correct with assumptions | Mostly correct | Incorrect |
| **Solution method** | Explained with bounds or gaps | Appropriate and explained | Appropriate | Wrong method |
| **Evaluation and validation** | Several runs, validated, plus sensitivity analysis | Several runs, validated | One run or no check | None |
| **Communication** | Clear memo and strong defence | Clear memo | Technical only | Unclear |

## Reading list


Dantzig, G. B. (1963). Linear Programming and Extensions. Princeton University Press.
The simplex method.

Bertsimas, D., & Tsitsiklis, J. N. (1997). Introduction to Linear Optimization. Athena Scientific.
Core LP and duality text.

Boyd, S., & Vandenberghe, L. (2004). Convex Optimization. Cambridge University Press.
Duality and convexity.

Land, A. H., & Doig, A. G. (1960). An automatic method of solving discrete programming problems. Econometrica, 28(3), 497–520.
Branch and bound.

Wolsey, L. A. (2020). Integer Programming (2nd ed.). Wiley.
Integer programming.

Kirkpatrick, S., Gelatt, C. D., & Vecchi, M. P. (1983). Optimization by simulated annealing. Science, 220(4598), 671–680.
Simulated annealing.

Williams, H. P. (2013). Model Building in Mathematical Programming (5th ed.). Wiley.
The modelling cycle.

Blum, W., & Leiß, D. (2007). How do students and teachers deal with modelling problems? In C. Haines et al. (Eds.), Mathematical Modelling (ICTMA 12): Education, Engineering and Economics (pp. 222–231). Horwood.
Modelling education research.